

Question: Considering a market vendor, with examples, discuss the concept of functions.

A **function** is a block of code that performs a specific task. Instead of writing the same code many times, you create a function once and use it whenever needed.

Example using a market vendor:

Think of a market vendor who sells fruits.

- A customer comes and asks for **1 kg of tomatoes**.
- The vendor weighs the tomatoes, collects the money, and gives the customer the tomatoes.
- Every time a customer wants tomatoes, the vendor repeats the same process.

This is similar to a **function** in programming. The function performs the same task whenever it is called.

Example 1

A function called sellTomatoes(quantity):

- Input: 2 kg
- Process: Weighs 2 kg of tomatoes.
- Output: Customer receives 2 kg of tomatoes.

Example 2

A function called calculateTotal(price, quantity):

- Price of tomatoes = UGX 3,000 per kg.
- Customer buys 2 kg.
- The function calculates:
 $3,000 \times 2 = \text{UGX } 6,000$.

Why functions are important

- They save time because you write the code once.
- They reduce repetition.
- They make programs easier to understand.
- They make it easier to fix mistakes because you only update the function once.

Conclusion:

Just as a market vendor follows the same steps whenever serving customers, a function performs the same task whenever it is called in a program. This makes programming simpler, faster, and more organized.